B.Tech CSE - Third Year | Lab 10

Assignment 10: Parent and Child Classes • *Aim: Use inheritance so child classes can access parent members.*

Experiment No. 10.1: Inherit sound() from Animal

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
class Animal {
    void sound() { System.out.println("Some sound"); }
}
class Dog extends Animal { }
public class Task_01_Inherit_sound_from_Animal {
    public static void main(String[] args) {
        Dog d = new Dog();
        d.sound();
    }
}
```

3. **Output:** `Some sound`
4. **Viva-Voce:** What is inheritance? What does extends mean? Can a child class use accessible parent methods?

Experiment No. 10.2: Call parent constructor with super(name)

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
class Person {
    String name;
    Person(String name) { this.name = name; }
}
class Student extends Person {
    Student(String name) { super(name); }
}
public class Task_02_Call_parent_constructor_with_supername {
    public static void main(String[] args) {
        Student s = new Student("Neha");
        System.out.println("Name: " + s.name);
    }
}
```

3. **Output:** `Name: Neha`
4. **Viva-Voce:** What does super(name) call? Why must a superclass constructor be called? What is a child class?

Experiment No. 10.3: Use inherited greet() in Teacher

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
class Person {
    String name;
    void greet() { System.out.println("Hello, " + name); }
}
class Teacher extends Person { }
public class Task_03_Use_inherited_greet_in_Teacher {
    public static void main(String[] args) {
        Teacher t = new Teacher();
        t.name = "Kiran";
        t.greet();
    }
}
```

3. **Output:** `Hello, Kiran`
4. **Viva-Voce:** What is an inherited member? What is the role of extends? How does greet() access name?